Lecture 13 - Attention and impulsivity

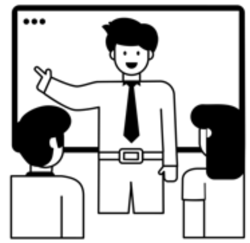
The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

Richard Ramsey

Today

Part 1

- Attention and impulsivity



Part 2

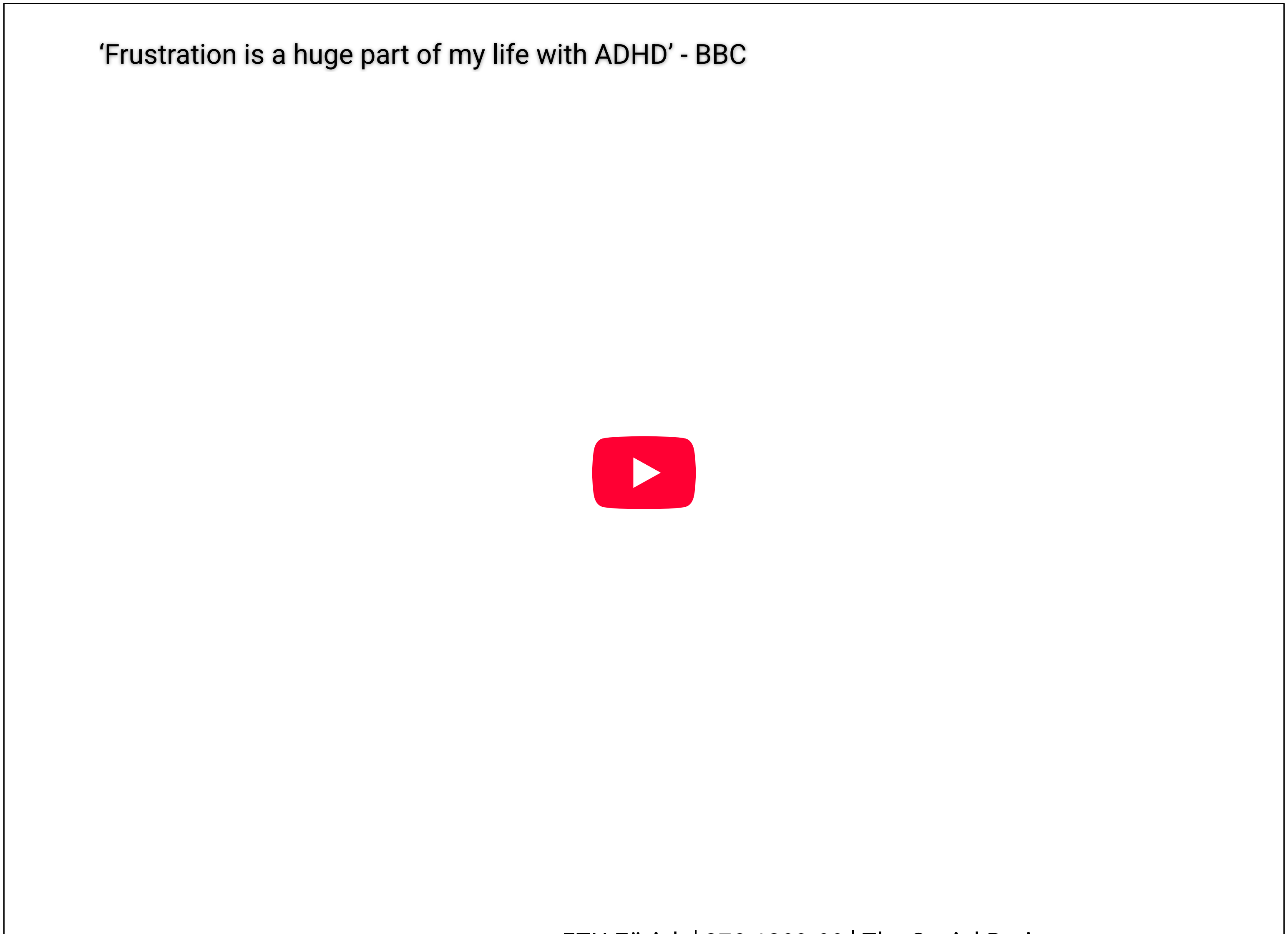
- Read articles and discuss

Overview

- Background & ADHD
- Performance on executive function tasks
  - Behavioural evidence
  - Neurobiological evidence

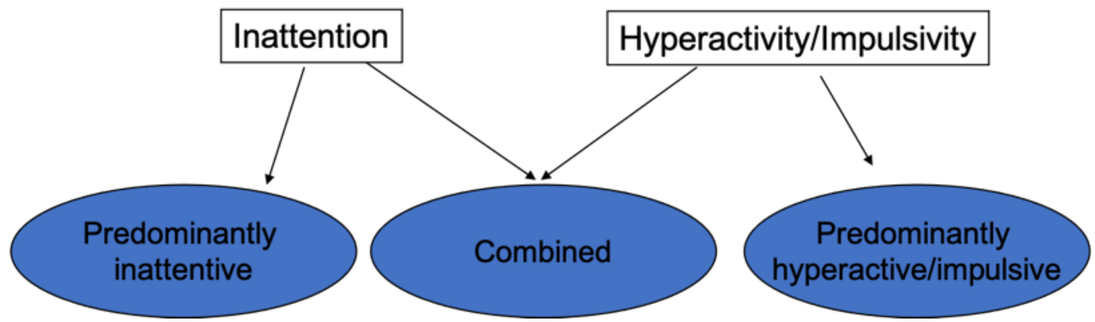
Background & ADHD

Living with ADHD



ADHD

- Attention Deficit/ Hyperactivity Disorder (ADHD) is defined by a combination of symptoms of
  - Inattention
  - Hyperactivity/ impulsivity
- 3 types:



Diagnosis

- Six (or more) of the following symptoms of inattention or hyperactivity-impulsivity have persisted for at least 6 months to a degree that is maladaptive and inconsistent with developmental level. Some examples as follows:
- Inattention
  - Often does not give close attention to details or makes careless mistakes in schoolwork.
  - Often has trouble keeping attention on tasks or play activities.
  - Often does not seem to listen when spoken to directly.
- Hyperactivity-impulsivity
  - Often fidgets with hands or feet or squirms in seat.
  - Often gets up from seat when remaining in seat is expected.
  - Often runs about or climbs when and where it is not appropriate.

Features

- Peak onset between the age of 3 and 4
- Affects 3%-5% children, prevalence higher in areas of socio-economic disadvantage
- Male/Female ratio
  - 4:1 for the hyperactivity/impulsivity type
  - 2:1 for the inattention type
- Co-morbidity: 50% of the children also show another psychiatric disorder:
  - Learning difficulties, Oppositional Defiant Disorder and Conduct Disorder, Anxiety and mood disorders, Substance abuse, Tic disorders
- Medical/Physical characteristics:
  - More accident prone, Sleep problems
  - Persists in adolescence for around 50-80% of individuals and in adulthood around 30-50% of individuals

Genetics

- Disorder clusters in families
  - Increased risk for 1st and 2nd order degree relatives (Faraone et al. 1995)
  - Siblings are 3 to 5 times more at risk (Biederman et al., 1992, Faraone et al., 1993)
  - Concordance is higher in monozygotic twins (50%-80%) than dizygotic twins (33%) (Bradley & Golden, 2001)

Several candidate genes (Durston & Konrad, 2007), some linked to dopaminergic system and others to the serotonergic system. Likely to have a complex genetic origin

Cold vs hot executive tasks

- “Cold” executive functions tasks are associated with working memory, inhibition and planning.
- “Hot” executive function tasks are associated with reward and reinforcement mechanisms

Stroop task

Baselines

GREEN RED BLUE

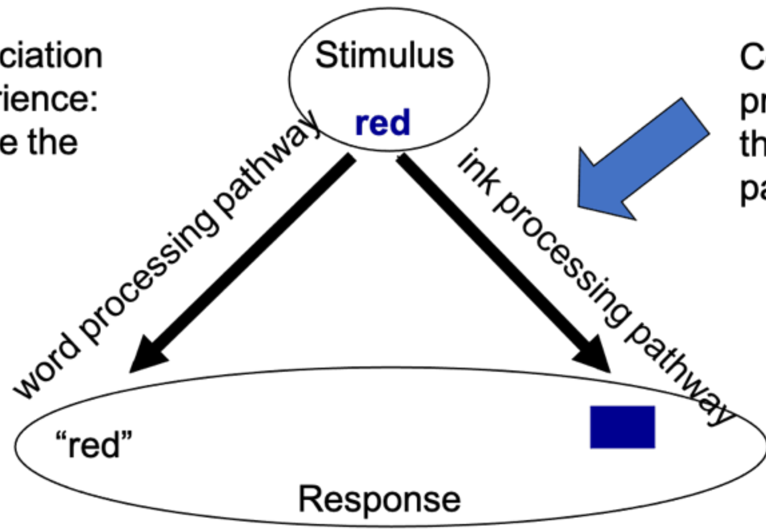
Cognitive control condition

BLUE GREEN RED

word is read automatically
➔ cognitive control is needed to inhibit reading response

Stroop task processes

stronger association through experience: easier to name the word



Cognitive control can provide a bias towards the less automatic pathway

if the task is to name the word
➔ little cognitive control

if the task is to name the ink colour
➔ more cognitive control

Delay aversion

- Greater preference for smaller-immediate over larger-delayed rewards (choice impulsivity)
- 1CHF now or 100 CHF in a year?
  - At which point do you shift your choice?
- Marshmallow test for kids. One now or two later?

Marshmallow test

The Marshmallow Test | Igniter Media | Church Video

Performance on executive function tasks

- Behavioural evidence
- fMRI evidence

Behavioural evidence

- Review of 8 meta-analyses:
  - Large effect: >0.50
  - Medium effect: 0.50-0.30
  - Small effect: <0.30

(Nigg, 2005)

Nigg 2005 results

Table 2. Selected Meta-analytic Findings in Neuropsychology of ADHD Versus Non-ADHD Children

Measure	Effect Size (*d*)
Spatial Working Memory (Spatial Span)	.75[a] to .85[b] to 1.14[b]
Response Suppression (Stop Task SSRT/SSRT Slope)	.61[a] to .64[c] to .94[d]
Signal Detection (CPT d-prime) Arousal	.72[e]
Stroop Naming Speed	.69[f]
Full Scale IQ	.61[g]
Set Shifting (Trails B Time)	.55[a] to .59[g] to 0.75[d]
Planning (Tower of London/Hanoi)	.51[a] to .69[a]
Mazes	.58[a]
Verbal Working Memory	.51[a] to .41[b]
Decision Speed on Go-Task	.49[c]
WCST Perseverations	.35[g]/.36[a] to .53[h]
Fluency	.27[d]
Stroop Interference	.25[f]
Covert Visual Spatial Orienting	.20[i]

Nigg 2005 results

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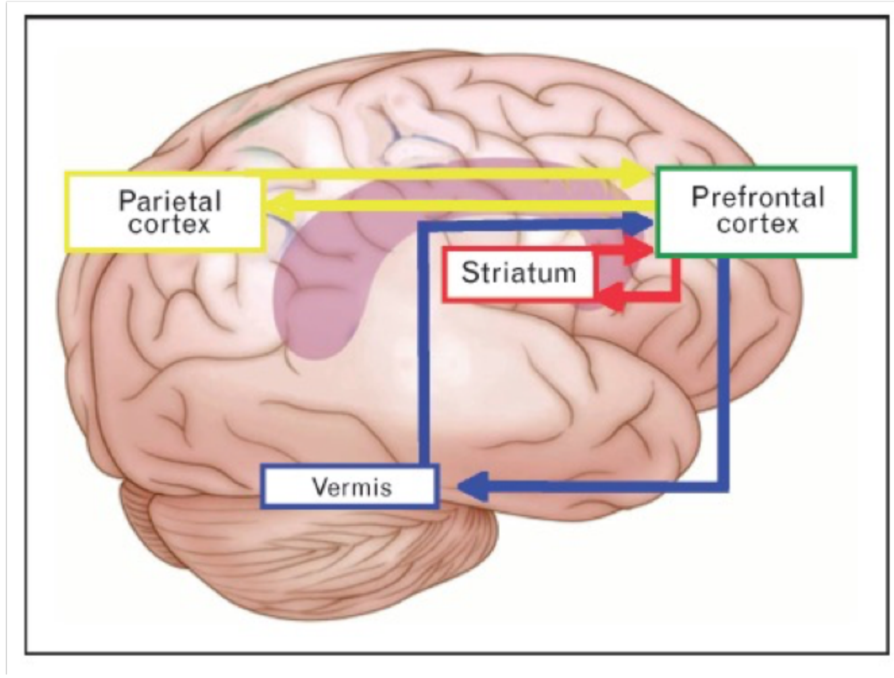
Neurobiological evidence

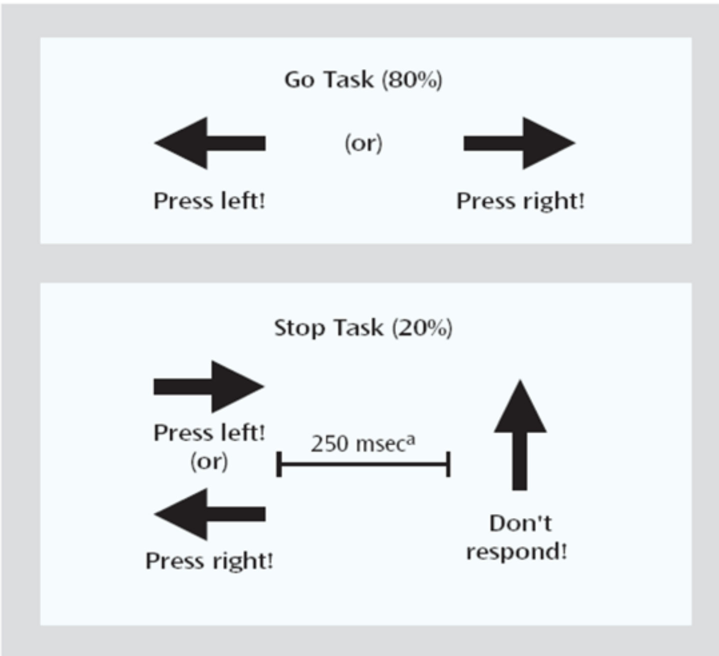
- Functional MRI studies reveal atypical activity in
  - Prefrontal cortex (esp. right inferior frontal)
  - Basal ganglia (esp. putamen)
  - Cerebellum
  - Parietal areas



(Casey & Durston, 2006; Rubia, 2018)

Response suppression task

Response suppression task

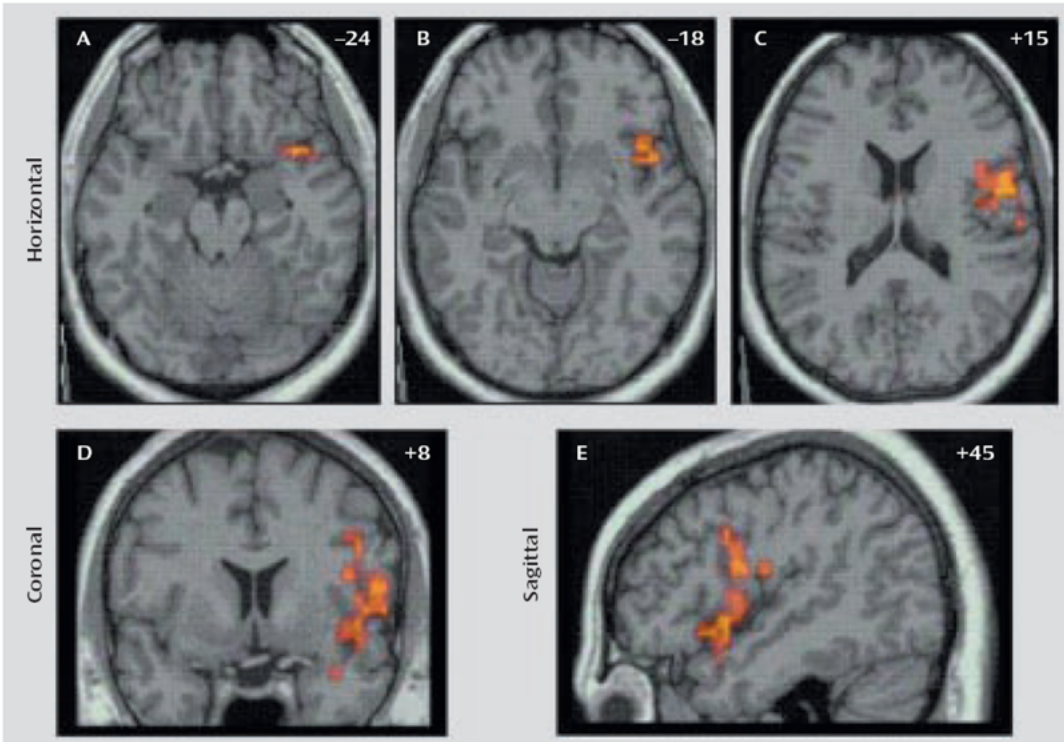


[a] The interval between horizontal and vertical arrows in the stop trials becomes smaller/larger in steps of 50 msec depending on each subject's performance to ensure 50% successful and 50% unsuccessful inhibition for each subject.

(Rubia et al., 2005; Rubia, 2018)

Response suppression results

FIGURE 2. Regions of Increased Activation During Successful Relative to Unsuccessful Inhibition in Healthy Subjects (N=21) Versus Medication-Naive Adolescents With ADHD (N=16)[a]



[a] Talairach coordinates are indicated for slice distance (in mm) from the intercommissural line for the horizontal (A, B, C) and sagittal slices (D, E). The focus reaches from the right orbitoprefrontal (see A) to inferior prefrontal cortex (B, C), deep into the insula (B, D, E) and precentral gyrus (C, E), and bordering the temporal cortex (A, B, D, E). Cluster-wise probability of false positive activation: p<0.01.

(Rubia et al., 2005; Rubia, 2018)

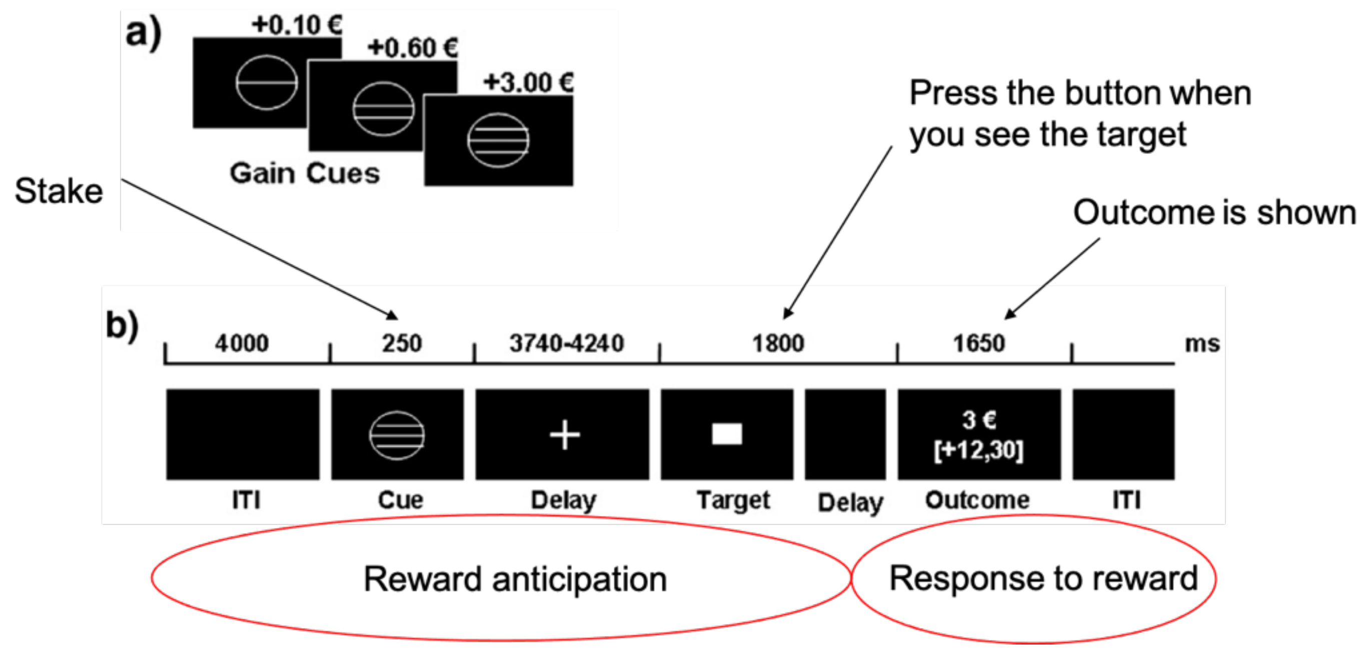
“Cold” task summary

Individuals with ADHD show:

- Atypical performance in EF tasks
- Atypical neural responses compared to controls in “cold” EF tasks
- What about “hot” tasks?

Reward task

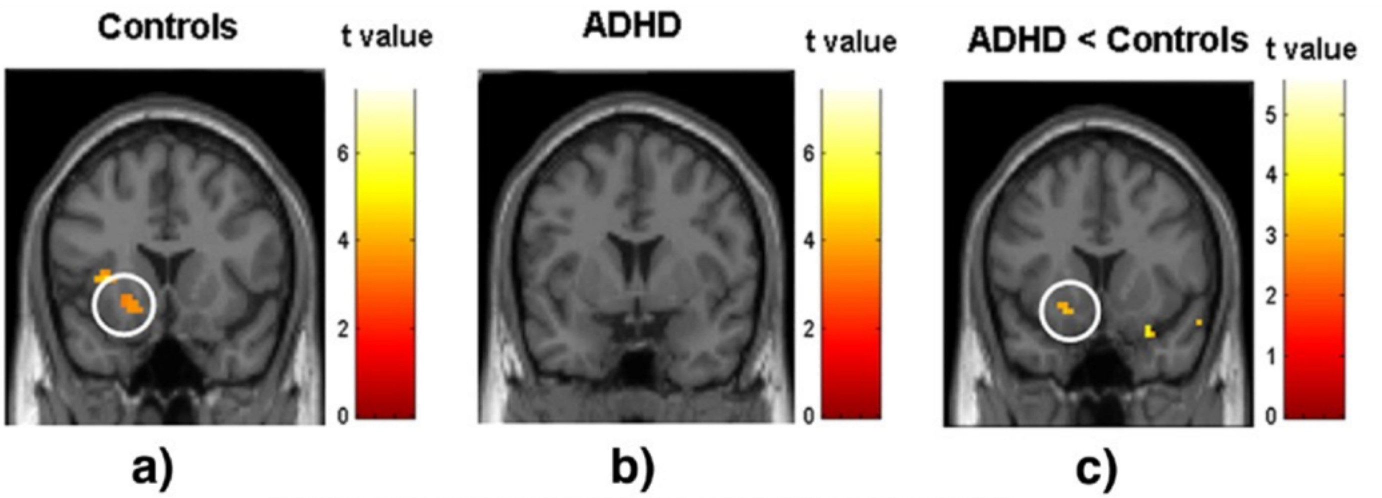
fMRI: Male adults with ADHD versus controls



(Ströhle et al., 2008)

Reward results

Reduced striatum activation when anticipating the gain

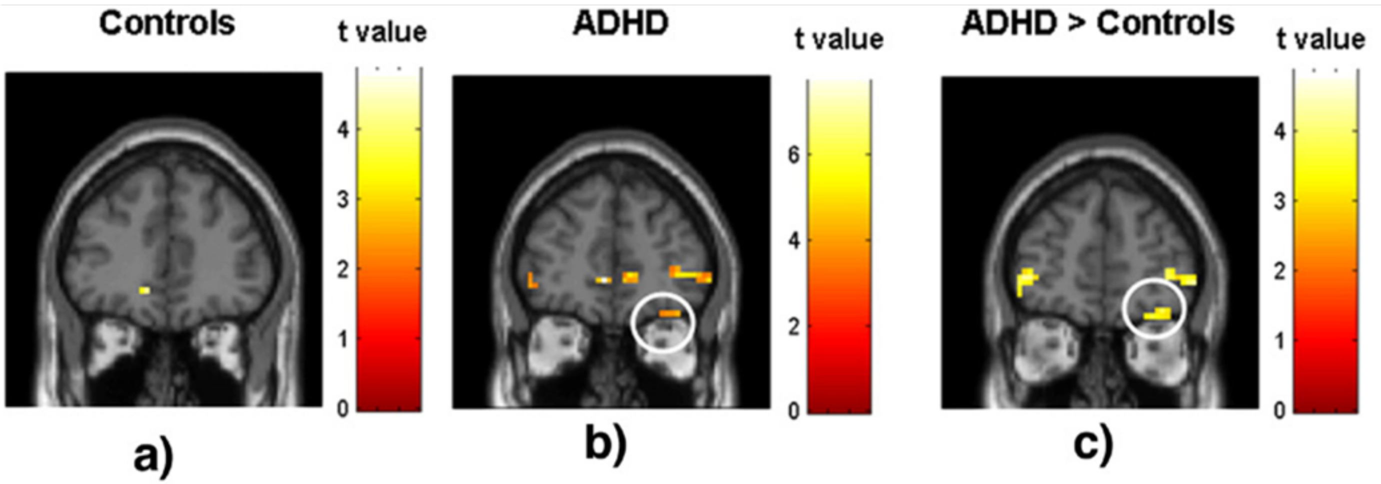


Ventral Striatal Activation during Anticipation

(Ströhle et al., 2008)

Reward results

Increased orbitofrontal activation when presented with gain outcome (over-sensitivity to reward)



Orbitofrontal Activation during Outcome

(Ströhle et al., 2008)

Summary

Individuals with ADHD show:

- Atypical performance in EF tasks
- Atypical neural responses compared to controls in “cold” and “hot” EF tasks

However…

Individual differences - Nigg 2005

% of ADHD children classified as impaired
➔ usually less than half!

Table 1. Illustrative Widely Used Neuropsychologic Measures Comparing ADHD (Combined Type) to Controls: Group Differences and Percent Impaired in 3 Samples

		Effect Size (d)			% ADHD Beyond Control 90th Percentile
Measure	Sample	d	η^2	*p*	
SSRT	MI (all)	.88	.133	<.001	51
	CO	.79	.101	<.001	45
RT Variability	MI	.75	.123	<.001	48
	CO	.77	.125	<.001	44
Stroop CW	MI	.50	.045	<.05	25
	CO	.84	.132	<.001	44
	MGH	.62	.09	<.001	25
CPT	MI	.91	.11	<.001	37
	CO	.54	.053	<.001	35
	MGH	.17	.01	.11	16
Trailmaking	MI	.35	.033	<.05	27
	CO	.35	.031	<.01	24

MI=Michigan sample
CO=Colorado sample
MGH=Massachusetts sample

Individual differences

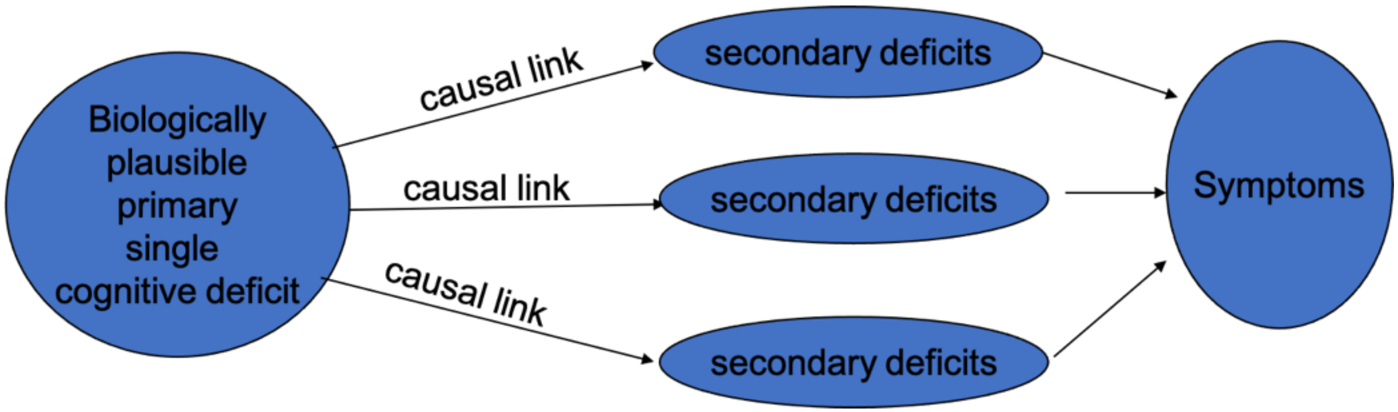
When ADHD children tested on both inhibition and delay aversion and looking at individual performances rather than group performance:

- 23% deficit on both
- 23% inhibition deficit only
- 15% delay aversion deficit only
- 39% no apparent deficit on any of the two

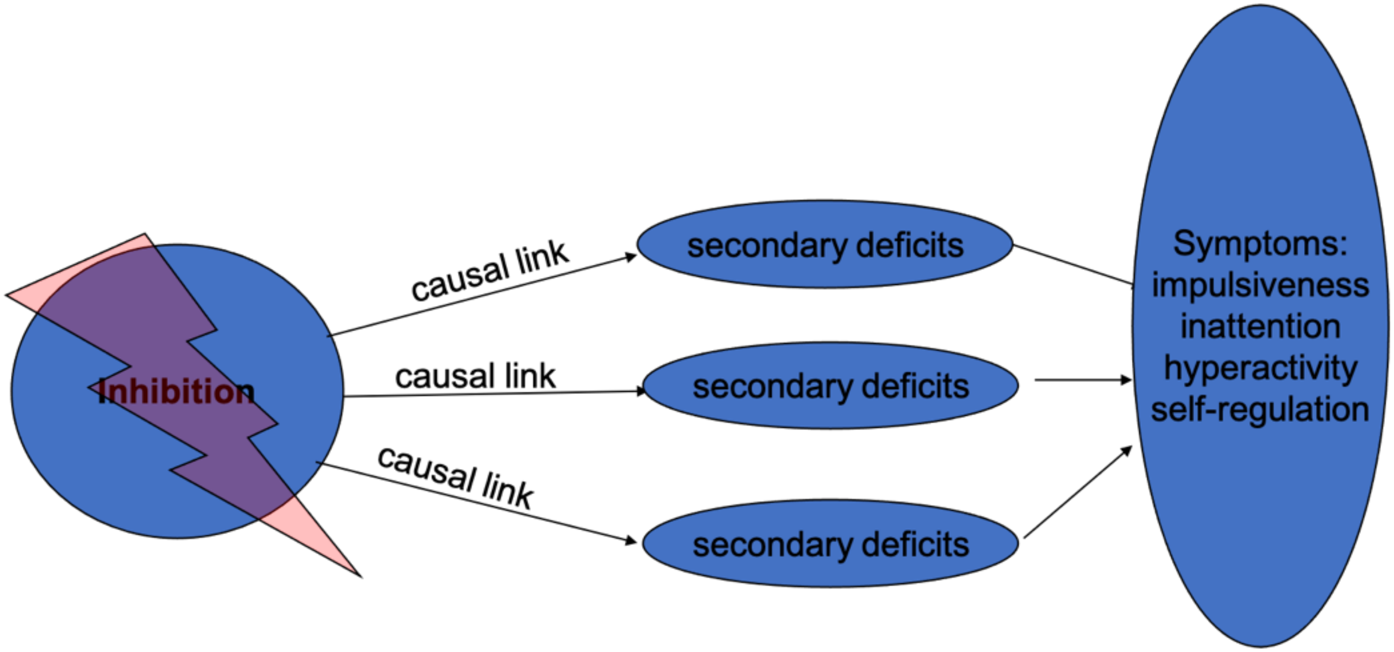
So what does this mean?

(Nigg, 2005)

Single cognitive mechanism?

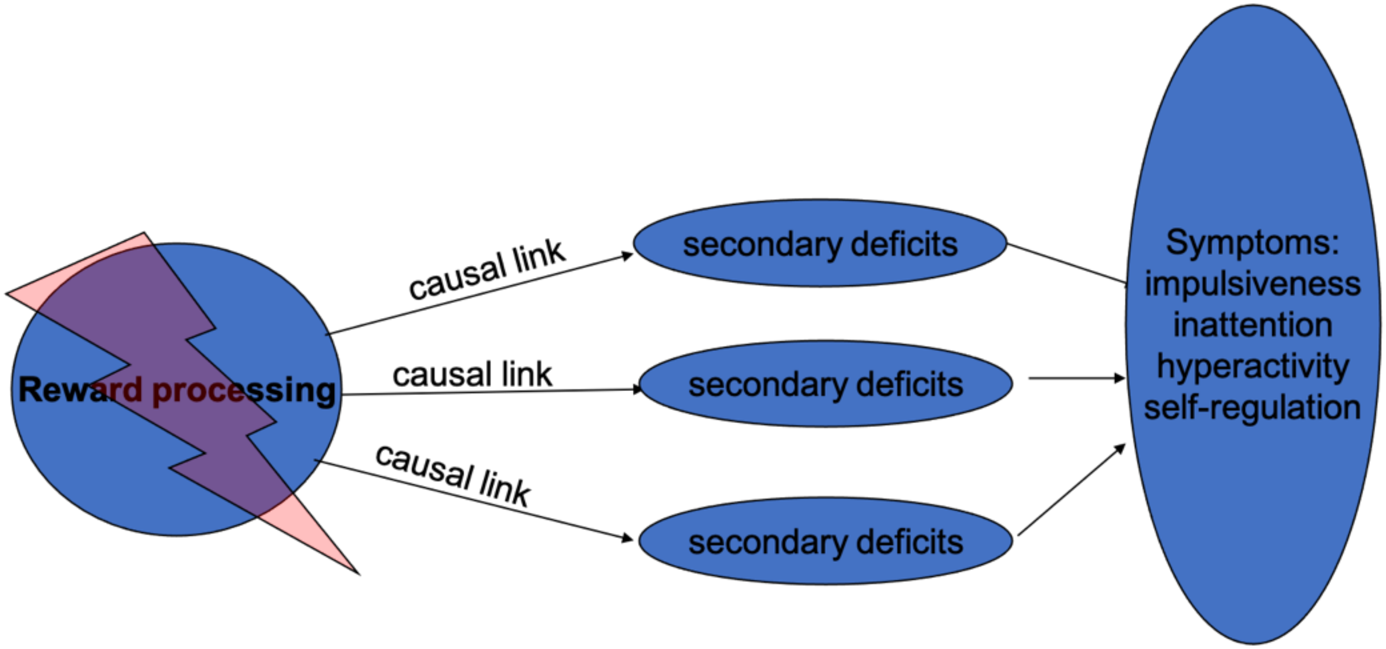


Inhibition



Inhibition
causal link
causal link
causal link
secondary deficits
secondary deficits
secondary deficits
Symptoms:
impulsiveness
inattention
hyperactivity
self-regulation

Reward


Reward processing
causal link
causal link
causal link
secondary deficits
secondary deficits
secondary deficits
Symptoms:
impulsiveness
inattention
hyperactivity
self-regulation

Future research

- Is there good evidence for a single cognitive deficit hypothesis?
- How well does a deficit in inhibition or motivation/reward processing explain the clinical symptoms associated with ADHD?
- Is it plausible to assume a single core deficit? What are the alternative options? And what are the implications for designing effective interventions?

(Castellanos et al., 2006; Rubia, 2018; Sonuga-Barke, 2005)

Take a break

Part 2 - Read and discuss

Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of the journal article: Rubia, 2018
- There is no need to read all of the paper. Pick a section of the paper that interests you and focus on that for the discussion.

References

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